SASWATA DAS

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Profile

I am a 3rd-year BS-MS Physics student at IISER Pune interested in observational and computational cosmology. I have used Python to select stellar-stream members from Gaia DR3, analyze IllustrisTNG simulation data, and implement MCMC samplers, and I want to apply statistical inference to dark-matter and cosmological constraints.

Research Interests

Observational and computational cosmology, dark-matter substructure, stellar streams and Galactic dynamics, astronomical survey data analysis, CMB data analysis, cosmological parameter inference, and statistical methods for astrophysics.

Education

BS-MS, Physics 2024 – 2029
Indian Institute of Science Education and Research (IISER) Pune
CGPA: 8.9 / 10

Research Projects

KSP 7.0 – Ripples in the Dark: Characterizing Stellar Streams as Dark Matter Substructure Probes Jun – Jul 2026
Indian Institute of Technology Bombay (IITB)
Mentors: Tamojeet Rowchoudhury (UC Berkeley), Ravi Kumar (University of Minnesota); Coordinator: Vihang Vidwans (IITB)
Code: github.com/Saswata-Das117/KSP-7.0-Stellar-Stream-Dark-Matter

- Built a **GD-1 stellar-stream member-selection pipeline** on **Gaia DR3**, combining astrometric, photometric, proper-motion, and theoretical isochrone cuts to isolate candidate stream stars.
- Transformed Gaia positions and proper motions into stream-aligned coordinates and characterized the proper-motion distribution of GD-1.
- Measured the stream's density structure using **kernel density estimation (KDE)** and stream-track fitting to search for gaps and underdensities that could signal dark-matter substructure.
- Analyzed **IllustrisTNG TNG50-1** simulation data to identify Milky Way-like host halos and quantify the dark-matter subhalos capable of perturbing streams.
- Reduced **3.01 million** Gaia DR3 sources to **722 candidate** GD-1 stream members using photometric, proper-motion, spatial-footprint, and density cuts; identified **two underdensities** at $\phi_1 \approx -42°$ and $-21°$ with significances of 2.7σ and 4.0σ.
- Propagated Gaia proper-motion uncertainties through the full selection with a **Monte Carlo** (100 realizations); recovered the $-21°$ gap in **99%** and the $-42°$ gap in **93%** of realizations, with membership varying by $\sim^{+35}_{-34}$ stars around a median of **643**.
- Extended the pipeline to **six other streams** (Indus, C-12, Ylgr, Orphan-Chenab, NGC 6397, Jhelum) with iterative sigma-clipped track fitting and multi-scale gap detection; found **13** candidate underdensities, the strongest at 7.9σ (Indus) and 7.5σ (C-12).
- Fitted stream ages by **length matching** (Fardal et al. 2015 particle spray in a Milky Way potential, following Lu et al. 2026), reproducing four published ages to within **0.02–0.29 Gyr**; derived new ages of **1.18 Gyr** (**0.54–2.58**) for Ylgr and **0.13 Gyr** (**0.07–0.22**) for C-12 over a progenitor-mass grid.

- Computed per-stream **subhalo encounter rates** across **103** TNG50-1 Milky Way analogues from orbit-derived shells, fitted ages, and per-subhalo relative speeds; Orphan-Chenab has the highest median rate (**0.032 per kpc**) with encounters in **98%** of halos, versus **6%** for C-12 and **1%** for NGC 6397.

Winter School: MCMC Methods Jan – Mar 2026
Indian Institute of Technology Roorkee (IITR)
Coordinator: Sumedh Gosavi (IITR)

- Implemented **Metropolis**, **Metropolis–Hastings** (with asymmetric proposals) and **Gibbs** samplers, alongside accept–reject and importance sampling, and studied the theory of Monte Carlo and Markov Chain Monte Carlo methods.
- Applied MCMC to Bayesian parameter estimation (e.g., coin-bias inference with credible intervals), examining posterior distributions and convergence using trace plots, burn-in and acceptance rates.
- Simulated the 2D Ising model across temperatures, and built a Metropolis-based 1D traffic-jam simulator as the final project, showing how a single slow car triggers backward-propagating congestion.

Capstone Project: Exoplanet Habitability Analysis May – Jul 2025
BRICS Astronomy 2025
Coordinator: Sally Macfarlane, Inter-University Institute for Data Intensive Astronomy (IDIA)

- Cleaned exoplanet datasets and performed feature selection and exploratory analysis using **NumPy**, **pandas**, **Matplotlib**, and **Seaborn** to study indicators of planetary habitability.
- Trained and evaluated **scikit-learn** classification models for habitability prediction.

Skills

Programming: Python
Libraries: NumPy, pandas, SciPy, Matplotlib, Seaborn, scikit-learn, Astropy, h5py, gala, gal-streams
Data and Simulations: Gaia DR3 (catalogue querying, coordinate transformations), IllustrisTNG
Methods: Statistical inference, MCMC, KDE-based density estimation, stellar-stream member selection, machine learning
Tools: Git, LaTeX

Achievements

- Awarded the **INSPIRE Scholarship** for academic excellence.
- Ranked in the **top 1%** in the ISC (Class XII) examinations.